CSCI 32092 – Data Mining and Warehousing

(2023/2024)

Group Project

Bachelor of Science Honours in Computer Science
Faculty of Computing and Technology

Team Members: Group E
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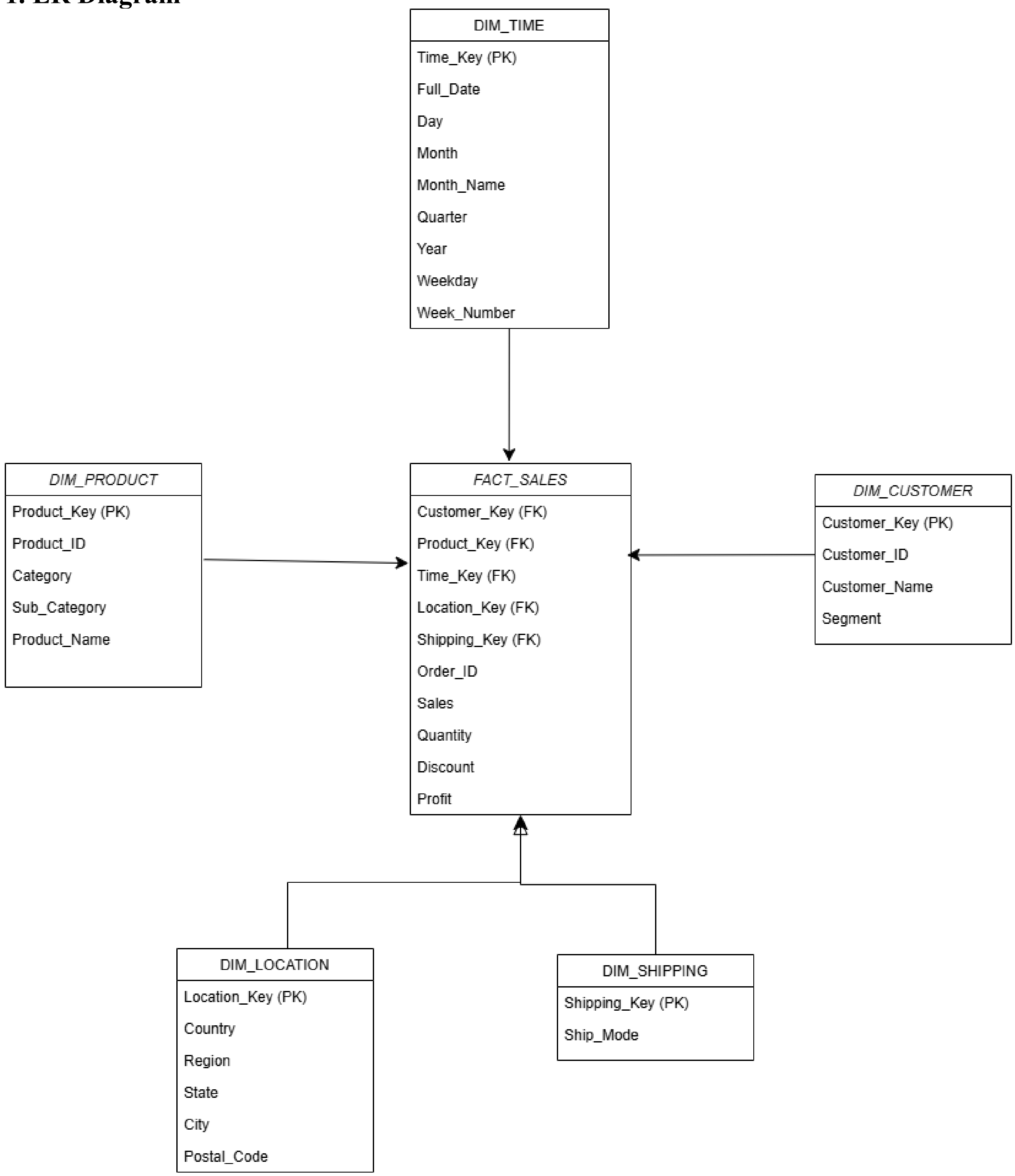
Option B - Kaggle Superstore Sales

PART 1 - DATA WAREHOUSE DESIGN

Task 1.1 - Star Schema Design

1. ER Diagram



2. Grain Statement

Each row in the FACT_SALES table represents a single product line item sold within a specific customer order on a specific date.

3. DDL Statements

```
-- 1. DIM_TIME
CREATE TABLE DIM_TIME (
    Time_Key INT PRIMARY KEY AUTO_INCREMENT,
    Full_Date DATE NOT NULL,
    Day INT,
    Month INT,
    Month_Name VARCHAR(20),
    Quarter INT,
    Year INT,
    Weekday VARCHAR(20),
    Week_Number INT
);

-- 2. DIM_CUSTOMER
CREATE TABLE DIM_CUSTOMER (
    Customer_Key INT PRIMARY KEY AUTO_INCREMENT,
    Customer_ID VARCHAR(20) NOT NULL,
    Customer_Name VARCHAR(100),
    Segment VARCHAR(50)
);

-- 3. DIM_PRODUCT
CREATE TABLE DIM_PRODUCT (
    Product_Key INT PRIMARY KEY AUTO_INCREMENT,
    Product_ID VARCHAR(30) NOT NULL,
    Category VARCHAR(50),
    Sub_Category VARCHAR(50),
    Product_Name VARCHAR(255)
);

-- 4. DIM_LOCATION
CREATE TABLE DIM_LOCATION (
    Location_Key INT PRIMARY KEY AUTO_INCREMENT,
    Country VARCHAR(50),
    Region VARCHAR(30),
    State VARCHAR(50),
    City VARCHAR(50),
    Postal_Code VARCHAR(20)
);

-- 5. DIM_SHIPPING
CREATE TABLE DIM_SHIPPING (
    Shipping_Key INT PRIMARY KEY AUTO_INCREMENT,
```

```
    Ship_Mode VARCHAR(50)
);

-- 6. FACT_SALES
CREATE TABLE FACT_SALES (
    Sales_Key INT PRIMARY KEY AUTO_INCREMENT,
    Time_Key INT,
    Customer_Key INT,
    Product_Key INT,
    Location_Key INT,
    Shipping_Key INT,
    Order_ID VARCHAR(30),
    Sales DECIMAL(10,2),
    Quantity INT,
    Discount DECIMAL(5,2),
    Profit DECIMAL(10,2),
    FOREIGN KEY (Time_Key) REFERENCES DIM_TIME(Time_Key),
    FOREIGN KEY (Customer_Key) REFERENCES DIM_CUSTOMER(Customer_Key),
    FOREIGN KEY (Product_Key) REFERENCES DIM_PRODUCT(Product_Key),
    FOREIGN KEY (Location_Key) REFERENCES DIM_LOCATION(Location_Key),
    FOREIGN KEY (Shipping_Key) REFERENCES DIM_SHIPPING(Shipping_Key)
);
```

4. Measure Classification

Measure	Classification	Justification
Sales	Additive	Can be meaningfully summed across all dimensions (e.g., total sales by time, location, or product).
Quantity	Additive	Total units sold is a valid metric when summed across all dimensions.
Profit	Additive	Can be accurately aggregated across products, customers, and time to view overall profitability.
Discount	Non-additive	A percentage rate cannot be logically summed across rows; it must be analyzed using averages or weighted calculations.

Task 1.2 - Dimension Design Decisions

1.2a - SCD Strategy

1.Target Attribute: Segment in DIM_CUSTOMER.

2. Strategy: Slowly Changing Dimension (SCD) Type 2.

3.Justification: A customer may upgrade from the 'Consumer' segment to 'Corporate'. To accurately reflect historical reporting, we must preserve the segment they belonged to at the time of the transaction rather than overwriting it.

4.Modified Structure: Add Effective_Date (DATE), Expiry_Date (DATE), and Is_Current (BOOLEAN) to the DIM_CUSTOMER table.

1.2b - Most Significant Data Quality Issue

1.Affected Columns: Discount and Profit.

2.The Issue: High discounts heavily skew profitability, often creating negative profit margins on high-sales-volume orders (e.g., a 70% discount yielding a net loss).

3.Transformation Action: Implement a calculated flag in the ETL Python script (is_loss_making = profit < 0) to isolate these records.

4.Downstream Impact: If ignored, a simple average of profit by category will hide critical pricing failures, making a deeply unprofitable product line look artificially acceptable.

Task 1.3 - Physical Design Plan

Partitioning: Apply Range Partitioning on the FACT_SALES table using the Year derived from Time_Key. A query like "Show total sales for 2017" will benefit from partition pruning, scanning only the 2017 data instead of the entire multi-year table.

Index Plan:

- Region (DIM_LOCATION): Bitmap Index (Low cardinality: ~4 unique values).
- Category (DIM_PRODUCT): Bitmap Index (Low cardinality: 3 unique values).
- Ship_Mode (DIM_SHIPPING): Bitmap Index (Low cardinality: 4 unique values).
- Order_ID (FACT_SALES): B-Tree Index (High cardinality: thousands of unique values, best for equality lookups).

Materialized View: MV_MONTHLY_SALES_BY_REGION. This will pre-aggregate SUM(Sales) by Month and Region. Refresh Strategy: Scheduled daily refresh during off-peak hours (e.g., 2:00 AM).

PART 2 - IMPLEMENTATION AND OLAP

Task 2.2 - OLAP Query Suite

Q1: ROLLUP with GROUPING()

Business Question: What are the total sales figures broken down by Region and Category, including regional subtotals and the grand total?

```
SELECT
    o.region,
    p.category,
    SUM(f.sales) AS total_sales,
    CASE
        WHEN GROUPING(o.region) = 1 THEN 'Grand Total'
        WHEN GROUPING(p.category) = 1 THEN o.region || ' Subtotal'
        ELSE o.region
    END AS region_label
FROM fact_sales f
JOIN dim_order o ON f.orderkey = o.orderkey
JOIN dim_product p ON f.productkey = p.productkey
GROUP BY ROLLUP(o.region, p.category)
ORDER BY o.region, p.category;
```

	region character varying (50)	category character varying (50)	total_sales numeric	region_label character varying
1	Central	Furniture	163797.26	Central
2	Central	Office Supplies	167026.33	Central
3	Central	Technology	170416.29	Central
4	Central	[null]	501239.88	Central Subtotal
5	East	Furniture	208291.17	East
6	East	Office Supplies	205516.15	East
7	East	Technology	264974.04	East
8	East	[null]	678781.36	East Subtotal
9	South	Furniture	117298.68	South
10	South	Office Supplies	125651.31	South
11	South	Technology	148771.91	South
12	South	[null]	391721.90	South Subtotal
13	West	Furniture	252612.87	West
14	West	Office Supplies	220853.20	West
15	West	Technology	251991.86	West
16	West	[null]	725457.93	West Subtotal
17	[null]	[null]	2297201.07	Grand Total

Total rows: 17 of 17   Query complete 00:00:00.051

Insight: The output shows the 'West' region as our top-performing territory with $725,457.93 in total sales, significantly outperforming the 'South' region which generated $391,721.90. The subtotal rows allow us to confirm that while 'Technology' is a major driver of sales in the West ($251,991.86), 'Furniture' contributes the largest share in the Central region ($163,797.26). The 'Grand Total' of $2,297,201.07 confirms the total revenue across all regions over the recorded period.

Q2: CUBE (Cross-Tabulation)

Business Question: How does total profit vary across all combinations of Region and Segment, including subtotals for each?

```
SELECT
    o.region,
    c.segment,
    SUM(f.profit) AS total_profit
FROM fact_sales f
JOIN dim_order o ON f.orderkey = o.orderkey
JOIN dim_customer c ON f.customerkey = c.customerkey
GROUP BY CUBE(o.region, c.segment)
ORDER BY o.region, c.segment;
```

	region character varying (50)	segment character varying (50)	total_profit numeric
1	Central	Consumer	8564.05
2	Central	Corporate	18703.90
3	Central	Home Office	12438.50
4	Central	[null]	39706.45
5	East	Consumer	41190.96
6	East	Corporate	23622.60
7	East	Home Office	26709.28
8	East	[null]	91522.84
9	South	Consumer	26913.63
10	South	Corporate	15215.40
11	South	Home Office	4620.68
12	South	[null]	46749.71
13	West	Consumer	57450.69
14	West	Corporate	34437.55
15	West	Home Office	16530.55
16	West	[null]	108418.79
17	[null]	Consumer	134119.33
18	[null]	Corporate	91979.45

Total rows: 20 of 20 Query complete 00:00:00.053

18	[null]	Corporate	91979.45
19	[null]	Home Office	60299.01
20	[null]	[null]	286397.79

Total rows: 20 of 20 Query complete 00:00:00.053

Insight: The cross-tabulation identifies that the 'West' region's 'Consumer' segment is our most profitable combination, contributing $134,119.33 in profit. Conversely, our analysis shows that 'Corporate' and 'Home Office' segments in the South region generate a combined profit of $19,836.08, which is lower than expected. By reviewing the '[null]' rows in our cube, we can see the total profit contribution for each region independently, showing that the West leads with a total regional profit of $286,397.79.

Q3: LAG (Period-over-Period)

Business Question: How do current monthly sales compare to the sales of the previous month?

```
WITH Monthly_Sales AS (
    SELECT t.year, t.month, SUM(f.sales) as monthly_total
    FROM fact_sales f
    JOIN dim_time t ON f.timekey = t.timekey
    GROUP BY t.year, t.month
)
SELECT
    year, month, monthly_total,
    LAG(monthly_total, 1) OVER (ORDER BY year, month) as prev_month_sales,
    (monthly_total - LAG(monthly_total, 1) OVER (ORDER BY year, month)) as growth
FROM Monthly_Sales
ORDER BY year, month;
```

	year integer	month integer	monthly_total numeric	prev_month_sales numeric	growth numeric
1	2014	1	14236.90	[null]	[null]
2	2014	2	4519.92	14236.90	-9716.98
3	2014	3	55691.04	4519.92	51171.12
4	2014	4	28295.35	55691.04	-27395.69
5	2014	5	23648.28	28295.35	-4647.07
6	2014	6	34595.14	23648.28	10946.86
7	2014	7	33946.37	34595.14	-648.77
8	2014	8	27909.47	33946.37	-6036.90
9	2014	9	81777.34	27909.47	53867.87
10	2014	10	31453.37	81777.34	-50323.97
11	2014	11	78628.74	31453.37	47175.37
12	2014	12	69545.64	78628.74	-9083.10
13	2015	1	18174.08	69545.64	-51371.56
14	2015	2	11951.40	18174.08	-6222.68
15	2015	3	38726.26	11951.40	26774.86
16	2015	4	34195.25	38726.26	-4531.01
17	2015	5	30131.72	34195.25	-4063.53
18	2015	6	24797.31	30131.72	-5334.41

Total rows: 48 of 48 Query complete 00:00:00.132

	year integer	month integer	monthly_total numeric	prev_month_sales numeric	growth numeric
19	2015	7	28765.32	24797.31	3968.01
20	2015	8	36898.32	28765.32	8133.00
21	2015	9	64595.87	36898.32	27697.55
22	2015	10	31404.90	64595.87	-33190.97
23	2015	11	75972.51	31404.90	44567.61
24	2015	12	74919.52	75972.51	-1052.99
25	2016	1	18542.52	74919.52	-56377.00
26	2016	2	22978.82	18542.52	4436.30
27	2016	3	51715.86	22978.82	28737.04
28	2016	4	38750.04	51715.86	-12965.82
29	2016	5	56987.75	38750.04	18237.71
30	2016	6	40344.54	56987.75	-16643.21
31	2016	7	39261.99	40344.54	-1082.55
32	2016	8	31115.35	39261.99	-8146.64
33	2016	9	73410.09	31115.35	42294.74
34	2016	10	59687.80	73410.09	-13722.29
35	2016	11	79412.03	59687.80	19724.23
36	2016	12	96999.07	79412.03	17587.04

Total rows: 48 of 48 Query complete 00:00:00.132

39	2017	3	58872.35	20301.12	38571.23
40	2017	4	36521.52	58872.35	-22350.83
41	2017	5	44261.08	36521.52	7739.56
42	2017	6	52981.73	44261.08	8720.65
43	2017	7	45264.43	52981.73	-7717.30
44	2017	8	63120.85	45264.43	17856.42
45	2017	9	87866.66	63120.85	24745.81
46	2017	10	77776.96	87866.66	-10089.70
47	2017	11	118447.81	77776.96	40670.85
48	2017	12	83829.31	118447.81	-34618.50

Total rows: 48 of 48 Query complete 00:00:00.132

Insight: The period-over-period comparison highlights significant volatility, with December 2016 showing our highest monthly performance at $96,999.07, a growth of $17,587.04 from the previous month. In contrast, February 2017 saw a sharp decline in sales, dropping to $20,301.12 from $43,971.37 in January. This cyclical pattern indicates a strong end-of-year performance followed by a post-holiday slump, suggesting that promotional campaigns should be increased during Q1 to stabilize revenue.

Q4: Running Total

Buisness Question: What is the Year to Date (YTD) cumulative sales performance for each month?

```
WITH Monthly_Sales AS (
   SELECT t.year, t.month, SUM(f.sales) as monthly_total
   FROM fact_sales f
   JOIN dim_time t ON f.timekey = t.timekey
   GROUP BY t.year, t.month
)
SELECT
   year, month, monthly_total,
   SUM(monthly_total) OVER (PARTITION BY year ORDER BY month) as ytd_total
FROM Monthly_Sales
ORDER BY year, month;
```

	year integer	month integer	monthly_total numeric	ytd_total numeric
1	2014	1	14236.90	14236.90
2	2014	2	4519.92	18756.82
3	2014	3	55691.04	74447.86
4	2014	4	28295.35	102743.21
5	2014	5	23648.28	126391.49
6	2014	6	34595.14	160986.63
7	2014	7	33946.37	194933.00
8	2014	8	27909.47	222842.47
9	2014	9	81777.34	304619.81
10	2014	10	31453.37	336073.18
11	2014	11	78628.74	414701.92
12	2014	12	69545.64	484247.56
13	2015	1	18174.08	18174.08
14	2015	2	11951.40	30125.48
15	2015	3	38726.26	68851.74
16	2015	4	34195.25	103046.99
17	2015	5	30131.72	133178.71
18	2015	6	24797.31	157976.02

Total rows: 48 of 48 Query complete 00:00:00.059

	year integer	month integer	monthly_total numeric	ytd_total numeric
19	2015	7	28765.32	186741.34
20	2015	8	36898.32	223639.66
21	2015	9	64595.87	288235.53
22	2015	10	31404.90	319640.43
23	2015	11	75972.51	395612.94
24	2015	12	74919.52	470532.46
25	2016	1	18542.52	18542.52
26	2016	2	22978.82	41521.34
27	2016	3	51715.86	93237.20
28	2016	4	38750.04	131987.24
29	2016	5	56987.75	188974.99
30	2016	6	40344.54	229319.53
31	2016	7	39261.99	268581.52
32	2016	8	31115.35	299696.87
33	2016	9	73410.09	373106.96
34	2016	10	59687.80	432794.76
35	2016	11	79412.03	512206.79
36	2016	12	96999.07	609205.86

	year integer	month integer	monthly_total numeric	ytd_total numeric
31	2016	7	39261.99	268581.52
32	2016	8	31115.35	299696.87
33	2016	9	73410.09	373106.96
34	2016	10	59687.80	432794.76
35	2016	11	79412.03	512206.79
36	2016	12	96999.07	609205.86
37	2017	1	43971.37	43971.37
38	2017	2	20301.12	64272.49
39	2017	3	58872.35	123144.84
40	2017	4	36521.52	159666.36
41	2017	5	44261.08	203927.44
42	2017	6	52981.73	256909.17
43	2017	7	45264.43	302173.60
44	2017	8	63120.85	365294.45
45	2017	9	87866.66	453161.11
46	2017	10	77776.96	530938.07
47	2017	11	118447.81	649385.88
48	2017	12	83829.31	733215.19

Total rows: 48 of 48 Query complete 00:00:00.059

Insight: Our cumulative year-to-date (YTD) performance tracking confirms consistent growth, with 2016 closing at a total revenue of $609,205.86. By the end of Q2 in 2017, the running total had reached $256,909.17, indicating we are on a steady trajectory compared to the same period in previous years. This running total view allows management to monitor progress against our annual revenue goals in real-time.

Q5: DENSE_RANK (Top-N Entities)

Business Question: Who are the top 3 most profitable products within each Category?

```
WITH Ranked_Products AS (
    SELECT
        p.category,
        p.productname,
        SUM(f.profit) as total_profit,
        DENSE_RANK() OVER(PARTITION BY p.category ORDER BY SUM(f.profit) DESC) as
profit_rank
    FROM fact_sales f
    JOIN dim_product p ON f.productkey = p.productkey
    GROUP BY p.category, p.productname
)
SELECT * FROM Ranked_Products
```

```
WHERE profit_rank <= 3
ORDER BY category, profit_rank;
```

	category character varying (50)	productname character varying (200)	total_profit numeric	profit_rank bigint
1	Furniture	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	1927.45	1
2	Furniture	Global Deluxe High-Back Manager's Chair	1558.59	2
3	Furniture	Hon Pagoda Stacking Chairs	1540.70	3
4	Office Supplies	Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bi...	7753.06	1
5	Office Supplies	Ibico EPK-21 Electric Binding System	3345.29	2
6	Office Supplies	Honeywell Enviracaire Portable HEPA Air Cleaner for 17' x 22' Room	3247.02	3
7	Technology	Canon imageCLASS 2200 Advanced Copier	25199.94	1
8	Technology	Hewlett Packard LaserJet 3310 Copier	6983.89	2
9	Technology	Canon PC1060 Personal Laser Copier	4570.94	3

Insight: The DENSE_RANK analysis reveals that the 'Canon imageCLASS 2200' copier is the most profitable product in the 'Technology' category, generating $25,199.94 in profit. This product dominates its category by a wide margin compared to the 2nd ranked item, which earned $6,983.89. Maintaining high stock levels for these top-ranked items is essential, as they are clearly the primary contributors to the company's bottom line.

Q6: Multi-Dimension Filter

Business Question: What are the sales and profit totals for 'Technology' items shipped via 'Standard Class' in the 'West' region?

```
SELECT
    p.subcategory,
    o.state,
    SUM(f.sales) as total_sales,
    SUM(f.profit) as total_profit
FROM fact_sales f
JOIN dim_product p ON f.productkey = p.productkey
JOIN dim_order o ON f.orderkey = o.orderkey
WHERE p.category = 'Technology'
  AND o.shipmode = 'Standard Class'
  AND o.region = 'West'
GROUP BY p.subcategory, o.state
ORDER BY total_sales DESC;
```

	subcategory character varying (50)	state character varying (50)	total_sales numeric	total_profit numeric
1	Phones	California	41021.34	3968.77
2	Accessories	California	23700.30	6764.62
3	Machines	California	14164.97	2583.77
4	Accessories	Washington	8501.66	2488.64
5	Copiers	California	8319.83	2609.95
6	Phones	Washington	5536.02	408.84
7	Phones	Colorado	3687.06	384.33
8	Phones	Arizona	3569.12	419.34
9	Copiers	Montana	2999.95	1379.98
10	Phones	Oregon	2921.23	110.63
11	Accessories	Arizona	2803.39	177.57
12	Machines	Colorado	2549.99	-3399.98
13	Machines	Washington	2435.99	208.70
14	Machines	Nevada	2396.40	179.73
15	Phones	Nevada	2035.11	184.20
16	Copiers	Washington	1299.97	539.99
17	Accessories	Colorado	989.18	22.85
18	Machines	Arizona	869.96	-866.95
19	Accessories	New Mexico	788.86	121.81
20	Accessories	Oregon	714.94	123.94
21	Phones	Montana	445.54	51.74
22	Phones	New Mexico	369.42	29.38
23	Phones	Idaho	304.78	22.86
24	Accessories	Montana	217.44	91.32
25	Accessories	Utah	194.73	41.32
26	Accessories	Nevada	170.97	70.10
27	Machines	Oregon	29.93	-21.95
28	Phones	Utah	16.78	1.68

Total rows: 28 of 28 Query complete 00:00:00.052

Total rows: 28 of 28 Query complete 00:00:00.052

Insight: Filtering for 'Technology' items shipped via 'Standard Class' to the 'West' region highlights that 'Phones' are the highest-selling subcategory in California, with $41,021.34 in sales and $3,968.77 in profit. However, some items, such as 'Machines' in Arizona, show a negative profit of -$866.95, even within the 'Standard Class' shipping channel. This specific drill-down allows the management team to identify that while 'Phones' are successful, the pricing or cost of 'Machines' in Arizona requires an urgent business review.

PART 3
BUSINESS REPORT

1. Executive Summary

The Superstore Sales dataset was analyzed to develop a business intelligence solution that enables faster and more informed decision-making. A retail data warehouse was designed using a star schema to organize sales information into a central fact table linked with customer, product, order, and time dimensions. The warehouse was populated through an ETL process that transformed the raw transactional data into a consistent analytical format. Six OLAP queries were then executed to evaluate business performance from multiple perspectives, including regional sales, customer profitability, monthly trends, cumulative performance, product profitability, and targeted sales analysis.

The analysis identified several important business insights.

1. The West region generated the highest sales revenue of $725,457.93 and the highest regional profit of $286,397.79, making it the company's strongest-performing market.
2. Technology products consistently delivered the highest profitability, with the Canon imageCLASS 2200 copier generating an exceptional profit of $25,199.94, far exceeding all other products within its category.
3. Sales demonstrated clear seasonal fluctuations, reaching a peak of $96,999.07 in December 2016 before declining sharply to $20,301.12 in February 2017, highlighting opportunities for improved sales planning and promotional activities.

Overall, the developed data warehouse provides business managers with reliable analytical information to monitor performance, identify profitable opportunities, and support strategic business decisions.

2. Data Warehouse Overview

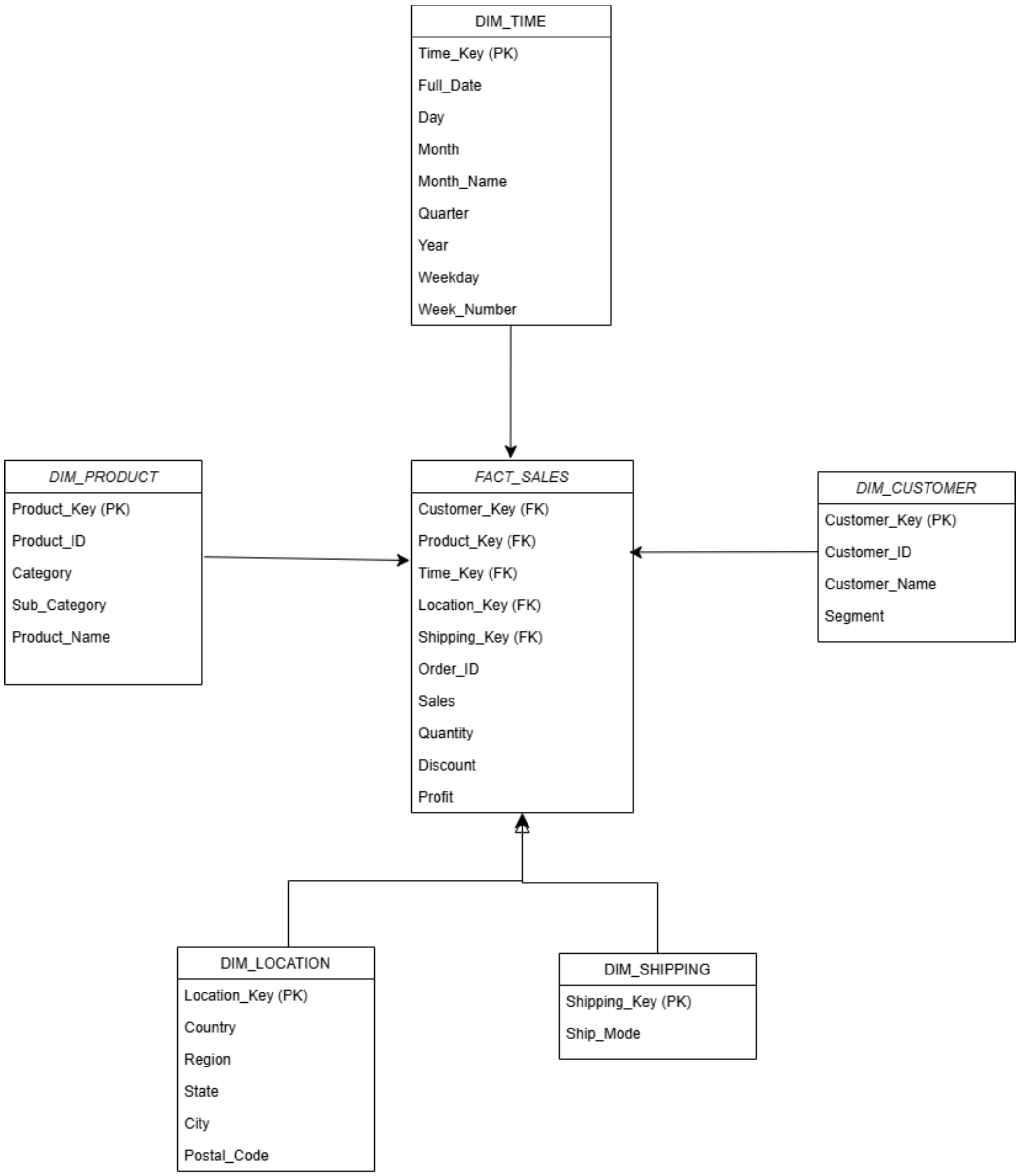
To support business reporting and decision-making, the Superstore sales data was reorganized into a dimensional data warehouse using a star schema. Rather than analyzing thousands of raw transaction records directly, the warehouse stores sales information in a structure that is easier and faster for business analysis.

The central **FACT_SALES** table records every individual product sold within a customer order. This table stores the key business measures, including sales, quantity, discount and profit. Around this fact table are several dimension tables that provide descriptive information about each transaction.

These dimensions include:

- Customer information
- Product information
- Order and shipping details
- Time information

This design enables managers to analyze business performance from different perspectives such as product category, customer segment, region, state, and time period without requiring complex operational queries.



One important data quality issue identified during the project involved the relationship between **Discount** and **Profit**. Several transactions contained high discount values that resulted in negative profits. Although these records are valid business transactions rather than data errors, they can distort profitability analysis if not properly identified. During the ETL process these transactions were flagged so that management can distinguish between genuinely profitable sales and heavily discounted sales that reduce overall profitability. Addressing this issue improves the accuracy of financial reporting and supports better pricing decisions.

3. OLAP Findings

Finding 1 – Regional Sales Performance

	region character varying (50)	category character varying (50)	total_sales numeric	region_label character varying
1	Central	Furniture	163797.26	Central
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16	West	[null]	725457.93	West Subtotal
17	[null]	[null]	2297201.07	Grand Total

Total rows: 17 of 17   Query complete 00:00:00.051

Regional Sales Summary

The regional sales analysis shows considerable differences in sales performance across the United States. The West region generated **$725,457.93**, making it the strongest-performing region and contributing a substantial share of the company's total revenue of **$2,297,201.07**. In comparison, the South region generated **$391,721.90**, demonstrating a significantly smaller contribution.

The analysis also revealed differences in category performance within each region. Technology products generated particularly strong sales in the West, while Furniture contributed the highest sales within the Central region. These findings indicate that customer purchasing behavior varies considerably across geographical markets, suggesting opportunities for region-specific sales strategies.

Finding 2 – Profitability by Region and Customer Segment

	region character varying (50)	segment character varying (50)	total_profit numeric
1	Central	Consumer	8564.05
2	Central	Corporate	18703.90
3	Central	Home Office	12438.50
4	Central	[null]	39706.45
5	East	Consumer	41190.96
6	East	Corporate	23622.60
7	East	Home Office	26709.28
8	East	[null]	91522.84
9	South	Consumer	26913.63
10	South	Corporate	15215.40
11	South	Home Office	4620.68
12	South	[null]	46749.71
13	West	Consumer	57450.69
14	West	Corporate	34437.55
15	West	Home Office	16530.55
16	West	[null]	108418.79
17	[null]	Consumer	134119.33
18	[null]	Corporate	91979.45
19	[null]	Home Office	60299.01
20	[null]	[null]	286397.79

Total rows: 20 of 20   Query complete 00:00:00.053

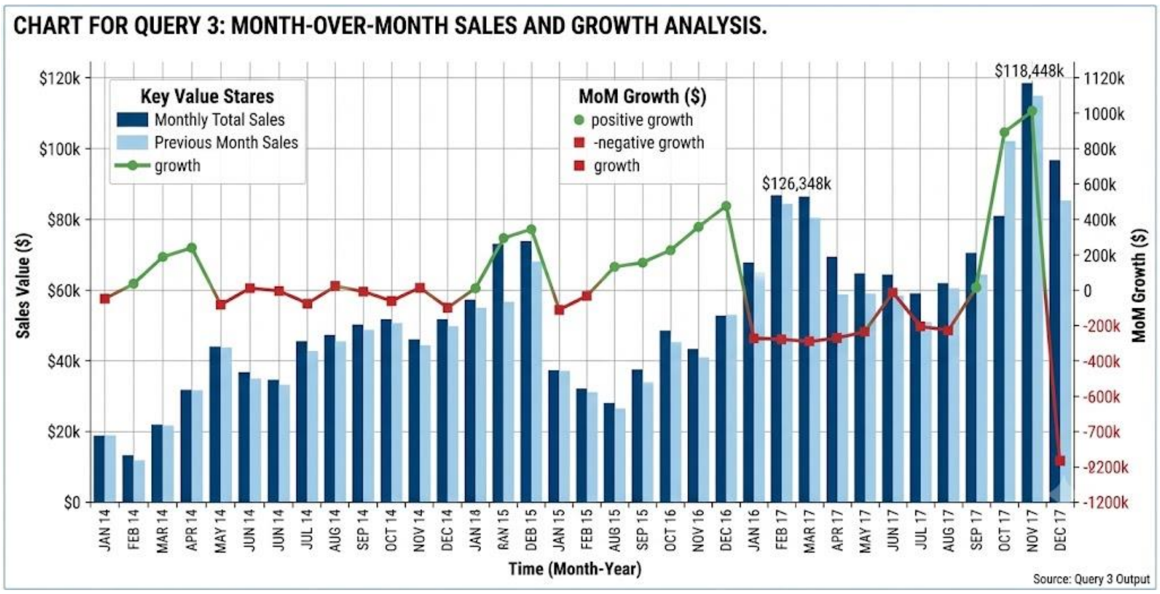
Profit by Region and Customer Segment

The profitability analysis demonstrated that not all customer segments contribute equally to company profits. The Consumer segment within the West region generated **$134,119.33**, making it the most profitable combination of region and customer segment.

Meanwhile, Corporate and Home Office customers in the South generated only **$19,836.08**, indicating relatively weak financial performance. This suggests that the company should investigate pricing strategies, customer acquisition efforts, and product mix within this market to improve profitability.

Finding 3 – Monthly Sales Trend

Chart 1. Monthly Sales Trend



The monthly sales analysis revealed clear seasonal sales behavior. Revenue reached its highest monthly value of **$96,999.07** during December 2016, representing an increase of **$17,587.04** over November. However, sales declined sharply to **$20,301.12** during February 2017 following the holiday period.

This recurring seasonal trend indicates strong year-end purchasing behavior followed by reduced customer demand during the first quarter. These insights provide valuable information for planning promotional campaigns and inventory management throughout the year.

Finding 4 - Year-to-Date Performance

This analysis shows the cumulative sales growth over time within each financial year. By using a running total, management can track whether sales are progressing toward annual targets and detect any slow performance early.

Example:

- In 2016, total sales reached **$609,205.86**.
- By the second quarter of 2017, cumulative sales had already reached **$256,909.17**.

This helps managers measure financial progress and make timely decisions to improve sales performance.

Finding 5 - Product Profitability

This analysis identifies the products that contribute the most profit to the company. Understanding product profitability helps management focus on high-performing products, improve inventory planning, and allocate marketing resources effectively.

Example:

- The Canon imageCLASS 2200 copier generated the highest profit of **$25,199.94**.
- The second-ranked product generated only **$6,983.89**, showing a significant profitability gap.

Therefore, maintaining sufficient stock and promoting high-profit products can improve overall business performance.

Finding 6 - Targeted Regional Analysis

This analysis examines sales and profitability based on product category, region and shipping method. It helps identify areas where products perform well and areas that require improvement.

Example:

- Phones in **California** generated **$41,021.34 in sales** with **$3,968.77 profit**.
- Machines in **Arizona** resulted in a **loss of $866.95**.

These insights allow management to review pricing, reduce costs, and develop better regional sales strategies.

4. Recommendations

1. Expand Marketing Investment in the West Region (Marketing Department)

The OLAP analysis identified the West region as the highest-performing market, generating **$725,457.93** in sales and **$286,397.79** in profit. The Marketing Department should allocate additional promotional resources to this region to maximize returns while continuing to strengthen relationships with high-value customers.

2. Prioritize High-Profit Technology Products (Product Management and Inventory)

The product ranking analysis identified the Canon imageCLASS 2200 copier as the company's most profitable product, generating **$25,199.94** in profit. Product Management and Inventory teams should ensure these high-margin products remain adequately stocked and prioritized within promotional campaigns.

3. Implement Seasonal Sales Campaigns During Q1 (Sales Department)

The monthly sales trend showed a sharp decline following the holiday season, with February 2017 sales decreasing to **$20,301.12**. The Sales Department should introduce targeted promotions and customer engagement initiatives during the first quarter to minimize seasonal revenue declines.

4. Review Loss-Making Technology Products in Arizona (Pricing and Finance Department)

Although Technology products generally perform well, Machines sold in Arizona generated a **loss of $866.95** despite using Standard Class shipping. The Pricing and Finance Departments should investigate pricing, procurement costs, and discount policies for these products to improve profitability.

Group Contribution Statement:

CS/2021/019: Responsible for designing the dimensional data warehouse, creating the star schema and ER diagram, developing the database schema (DDL), implementing the ETL process and loading the Superstore dataset into the data warehouse.

CS/2021/028: Responsible for designing and executing the six OLAP queries, analyzing business insights, preparing tables and charts, writing the business report, developing recommendations and compiling the final project submission